

ALSu_DCCT_Testing

This repository contains an automated Python-based testing suite for ALSu DCCT (Direct Current Current Transformer) Modules. This suite controls various lab instruments via PyVISA to perform rigorous fault and current tests, analyzes the acquired waveforms, and automatically generates comprehensive PDF test reports.

Features

- **FLT12 Fault Testing:** Tests the FLT12 fault output status (an open emitter output) by decrementing the -15V supply in 250mV steps and measuring the voltage to verify assertion and de-assertion thresholds.
- **AC Current Testing:** Drives an AC signal into a Kepco supply to push current through the DCCT. It verifies that Channel 1 and Channel 3 are matched in phase, Channel 2 is 180 degrees out of phase, and output signals meet valid amplitude levels.
- **Waveform Analysis:** Unpacks ADC waveform data from the oscilloscope and calculates frequency and phase shift using Fast Fourier Transform (FFT).
- **Automated PDF Reporting:** Generates a complete PDF report with ReportLab, detailing pass/fail statuses, technician demographics, and embedded waveform plots.
- **Modular Instrument Control:** Uses PyVISA utilities to connect to and control instruments via USB or Ethernet/IP connections.

Hardware Requirements

The testing suite is designed to interface with the following lab equipment:

- **Power Supply:** Rigol DP800 Series (e.g., DP832, DP831, DP822, DP811).
- **Signal Generator:** Rigol DG4000 Series.
- **Oscilloscope:** Tektronix DPO4000 or MSO4000 Series.
- **Digital Multimeter (DMM):** Keysight 34461A or Keithley 2100 Series.
- **Additional PSU:** Kepco Power Supply (driven by the signal generator for AC current testing).
- **Other Equipment:** Other equipment may be substituted by creating a new instrument library using identical commands to others in the library, for ease of compatibility. See, for example, switching between the Keysight 34461A vs Keithley 2100 DMMs seamlessly by changing only the address and the DMM object instantiation.

Software Requirements

- Python 3.x
- Dependencies listed in `requirements.txt` :
 - `PyVISA==1.14.1`
 - `reportlab==4.3.1`
 - `matplotlib==3.10.1`
 - `numpy==2.2.4`
 - (See `requirements.txt` for the full list of required packages)

Repository Structure

The project is organized into modular scripts separating instrument drivers, testing logic, and data reporting:

```

ALSu_DCCT_Testing/
├── main.py                # Main execution script and orchestrator
├── plotter_calculator.py  # Waveform unpacking, math (FFT), and matplotlib plotting
├── report_generator.py     # PDF generation using ReportLab
├── requirements.txt        # Python package dependencies
├── .gitignore              # Git ignore rules
├── README.md               # Project documentation
├── functional_tests/       # Test execution modules
│   ├── fault_test.py       # FLT12 Fault 1 and 2 test logic
│   └── current_test.py     # AC current waveform test logic
├── instrument_modules/     # PyVISA instrument drivers
│   ├── keithley_2100.py     # Keithley DMM driver
│   ├── keysight_34461a.py  # Keysight DMM driver
│   ├── rigol_dg4000.py     # Rigol Signal Gen driver
│   ├── rigol_dp800.py      # Rigol PSU driver
│   ├── Tek_DPO4000.py      # Tektronix Oscilloscope driver
│   └── visa_utils.py       # VISA connection utilities
├── Test_Data/              # Dynamically generated root directory for test artifacts
│   └── DCCT_<SN>-<Timestamp>/ # Unique test instance folder
│       ├── DCCT_<SN>_Report.pdf # Final generated report
│       └── raw_data/         # Subdirectory for raw test data
│           ├── *.csv         # Technician info, raw channel data, fault data
│           └── *.png         # Saved waveform plots

```

Installation

1. Clone the repository:

```

git clone [https://github.com/capotostobnl/ALSu_DCCT_Testing.git]
(https://github.com/capotostobnl/ALSu_DCCT_Testing.git)
cd ALSu_DCCT_Testing

```

bash

2. Create and activate a virtual environment (optional but recommended):

```

python -m venv .venv
source .venv/bin/activate # On Windows use: .venv\Scripts\activate

```

bash

3. Install the required Python packages:

```
pip install -r requirements.txt
```

bash

Configuration

Before running the tests, ensure your instrument IP addresses and USB VISA resource strings are correctly configured at the top of `main.py` :

```
# *****  
# *****Set Instrument IP Addresses*****  
PSU_IP_ADDRESS = "10.0.142.1" # Set PSU IP Address here  
SIG_GEN_IP_ADDRESS = "10.0.142.2" # Set signal generator IP Address here  
SCOPE_IP_ADDRESS = "10.0.142.3" # Set scope IP Address  
# DMM_ADDRESS = "USB0::0x05E6::0x2100::8020357::INSTR"  
DMM_ADDRESS = "10.0.143.23" # Set DMM Address here  
# *****
```

python

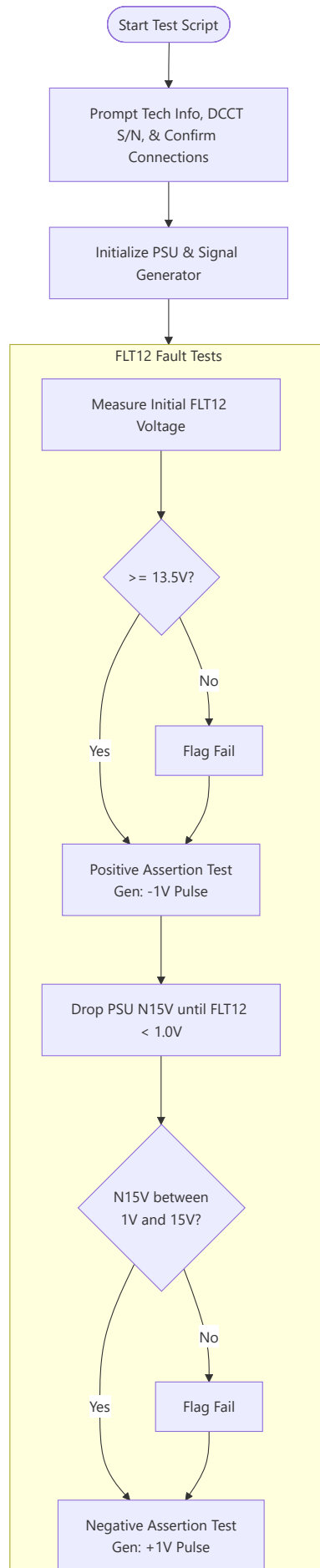
If needed, change object instantiation below the address settings, as well.

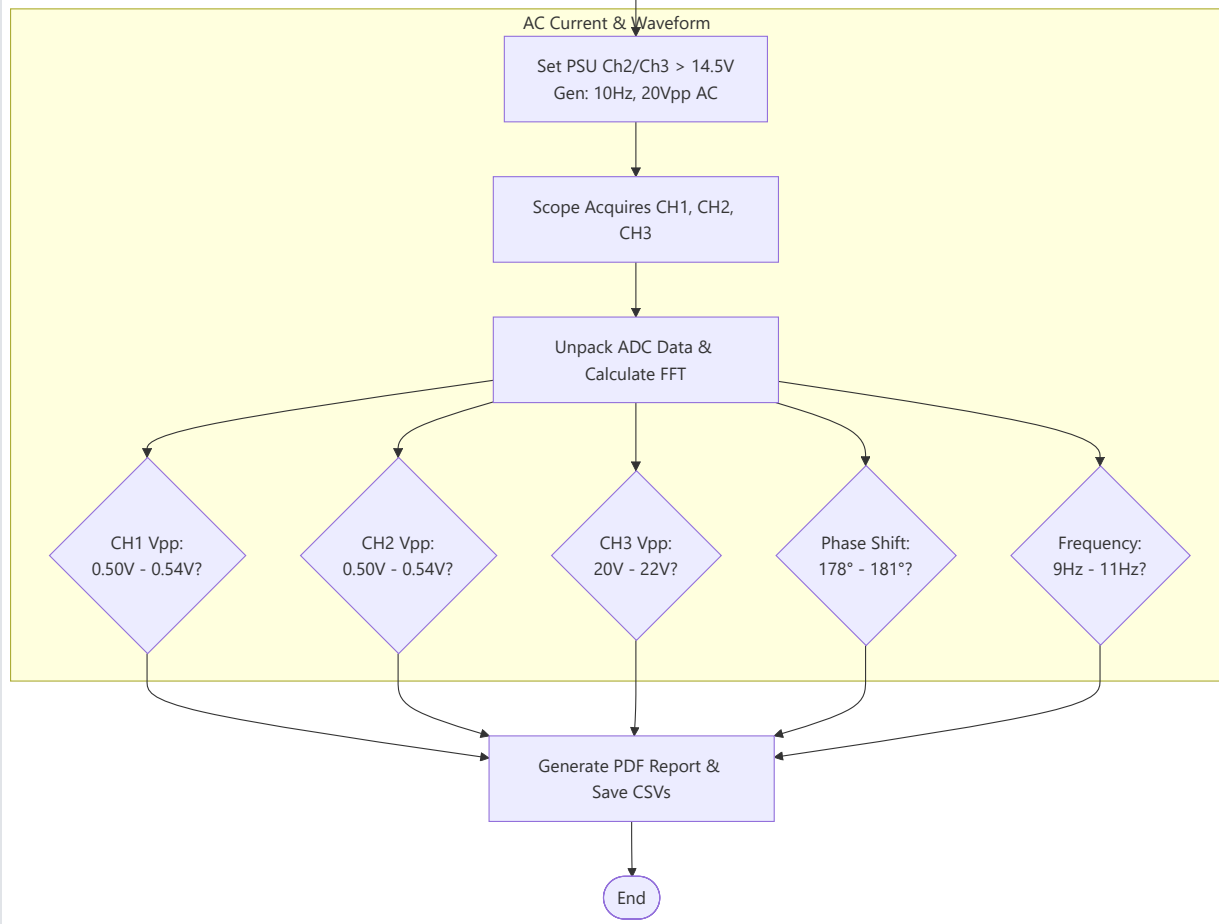
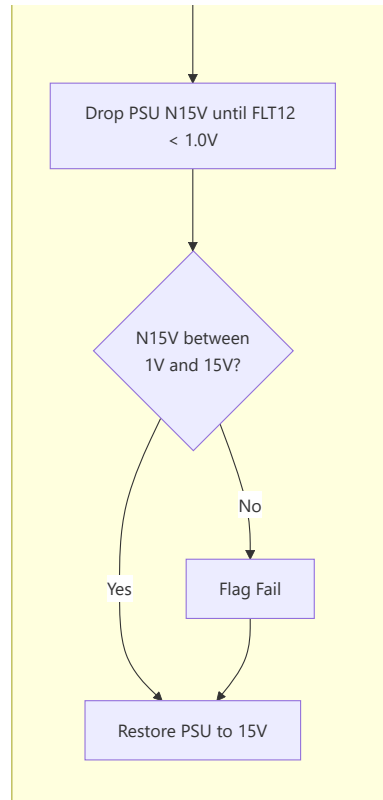
```
# *****  
# *****Create Instrument Objects*****  
  
psu = DP800(connection_method="IP", address=PSU_IP_ADDRESS)  
gen = DG4000(connection_method="IP", address=SIG_GEN_IP_ADDRESS)  
scope = DPO4000(connection_method="IP", address=SCOPE_IP_ADDRESS)  
# dmm = Keithley2100(connection_method="USB", address=DMM_ADDRESS)  
dmm = Keysight34461A(connection_method="IP", address=DMM_ADDRESS)  
# *****
```

python

Testing Process Flow & Pass/Fail Parameters

The script follows a strictly defined sequence of initializations, fault threshold verifications, and AC waveform validations. Below is a flowchart detailing this process and the exact pass/fail parameters evaluated at each step.





Usage

1. Power on all required instruments (Kepco PSU, Rigol PSU, DMM, Rigol function generator, and Tek Oscilloscope).
2. Ensure local network or USB connections to instruments are established.
3. Run the main testing script:

```
python main.py
```

```
bash
```

4. Follow the interactive prompts in the console to:
 - Enter your name and Life # (technician data).
 - Enter the DCCT Serial Number (scan barcode).
 - Confirm wiring connections via on-screen prompts (e.g., ensuring GRAY CT conductors go to GRAY terminal blocks, and BLUE conductors go to BLUE terminal blocks).
 - Confirm instrument setups (e.g., differential probes set to x20 Attenuation).
 - Refer to the 'George Ganetis' diagram, as needed, for reference.
5. The script will automatically run the FLT12 Fault 1/Fault 2 tests, followed by the DCCT Current tests.
6. Upon completion, a PDF report will be generated and saved in the `Test_Data/DCCT_<SN>-<Timestamp>` directory, and will open automatically on your machine. Raw test data (`.csv` files) and waveform plots (`.png` files) are also saved in a `raw_data` subdirectory for archival purposes.

ALSu DCCT Test Setup & Procedure Checklist

This checklist guides the technician through the physical setup, instrument configuration, and software execution required to test the ALSu DCCT modules.

1. Pre-Test Preparation

- ☐ The testing laptop is connected to the local instrument network (ensure IPs `10.0.142.x` and `10.0.143.x` are reachable).
- ☐ The Python virtual environment is activated.
- ☐ Required packages from `requirements.txt` are installed.

2. Hardware Power-Up

Ensure all laboratory instruments are turned ON:

- ☐ Kepco Power Supply
- ☐ Rigol DP800 Series Power Supply
- ☐ Rigol DG4000 Series Signal Generator
- ☐ Tektronix DPO4000/MSO4000 Series Oscilloscope
- ☐ Keysight 34461A or Keithley 2100 Digital Multimeter (DMM)
- ☐ Differential Probes for Oscilloscope

3. Physical Connections & Wiring

- ☐ **GRAY** CT conductors are routed and secured to the **GRAY** terminal blocks.
- ☐ **BLUE** conductors are routed and secured to the **BLUE** terminal blocks.
- ☐ Conductors loop through the DCCT only once, and wire is at least 2" away from rear of DCCT Chassis (Past the yellow spacer sticks)

4. Manual Instrument Configuration

While the script handles most parameters, the following must be set manually prior to execution:

- ☐ **Kepco PSU Front Panel Configuration:**
 - ☐ VOLTAGE CONTROL Switch: **OFF**
 - ☐ CURRENT CONTROL Switch: **OFF**
 - ☐ *Note: MODE Switch and Potentiometer positions do not matter.*

- ☐ **Oscilloscope Probes:**
 - ☐ Differential probes are powered **ON**.
 - ☐ Differential probes are physically set to **x20 Attenuation**.

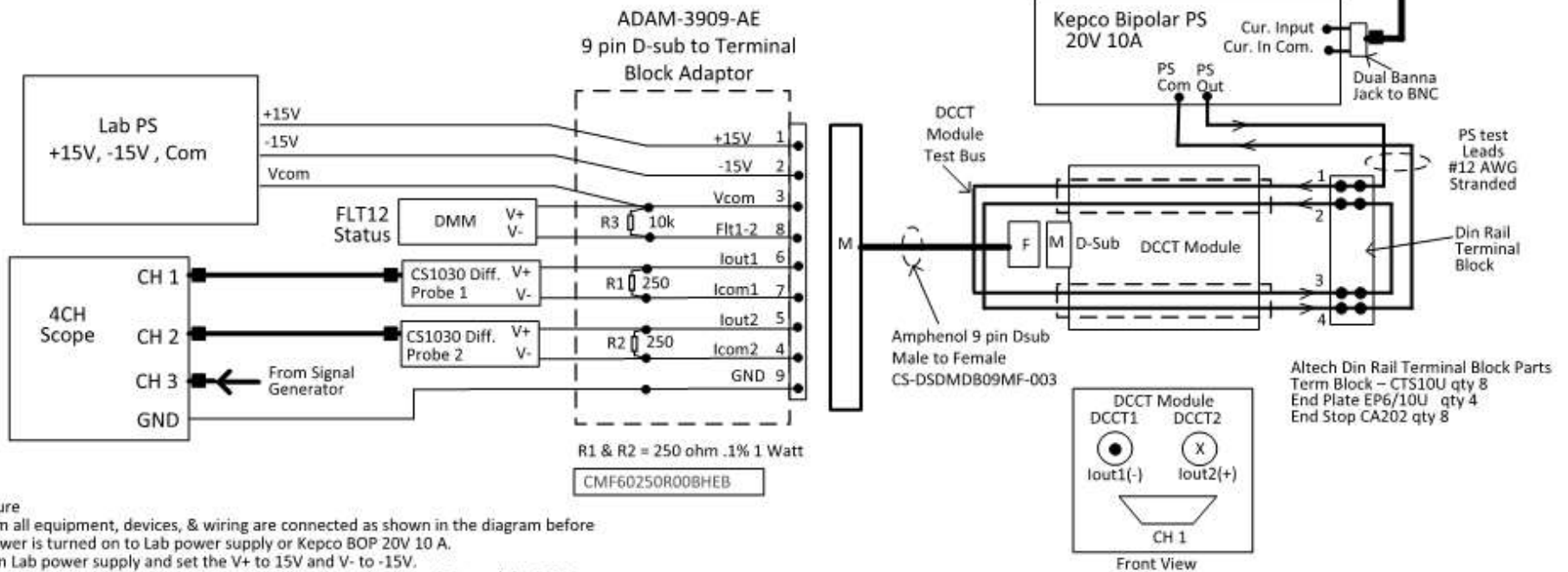
5. Software Execution

- ☐ Open a terminal/command prompt and run: `python main.py` .
- ☐ When prompted, enter your **Name** and confirm.
- ☐ When prompted, enter your **Life #** and confirm.
- ☐ Scan or manually enter the **DCCT Serial Number** (the script will capture the first 13 characters) and confirm.
- ☐ Acknowledge the on-screen connection confirmation prompts by pressing `Return` .

6. Testing & Review

- ☐ Allow the script to initialize the instruments and run the **FLT12 Fault 1 & 2 Tests**.
- ☐ If FLT12 tests fail, select whether to retry, restart with a new unit, or proceed to the next step.
- ☐ Allow the script to run the **DCCT Current Tests** (Phase, Frequency, and Voltage validation).
- ☐ Once completed, verify that the automated **PDF Test Report** opens on your screen.
- ☐ Verify that raw `.csv` and `.png` plot data were successfully saved in the new `Test_Data/DCCT_<SN>-<Timestamp>/raw_data/` directory.
- ☐ Affix a Green or Red 'QA' sticker to indicate the unit's pass or fail status
- ☐ Print reports as needed, and upload to Git repository at end of shift for archival

DCCT Module Test System GG 2/28/2025



Test Procedure

- Confirm all equipment, devices, & wiring are connected as shown in the diagram before any power is turned on to Lab power supply or Kepco BOP 20V 10 A.
- Turn on Lab power supply and set the V+ to 15V and V- to -15V.
- Confirm that the DMM reading above 14 Vdc on FLT12 status signal. Record the DMM reading in traveler.
- Set the Kepco power supply output + 1 amp DC.
- Lower the -15V on the Lab power supply until the FLT12 status signal on the DMM goes below 1 Vdc. Record DMM reading in traveler. Restore the -15V on the Lab power supply.
- Set the Kepco power supply output - 1 amp DC.
- Lower the -15V on the Lab power supply until the FLT12 status signal on the DMM goes below 1 Vdc. Record DMM reading in traveler. Restore the -15V on the Lab power supply.
- Setup Scope CH 1 & 2 are set to 0.1 V/Div. and 50 msec/Div. for the two probes and CH 3 is set to 10 V/ Div.
- Turn on signal generator and set it up for sin wave with an amplitude of +10 V to -10 V at a 10 Hz frequency. Use the scope CH 3 to confirm the waveform amplitude.
- Turn on the two probes and make sure they are connected as shown in the diagram.
- Turn on the Kepco power supply and confirm the CH1 & CH2 have a sine wave signals with the CH21 in opposite polarity of CH1 & CH3.
- Take a Snapshot on the scope and transfer data to traveler.
- Record DCCT module serial number on traveler.
- Turn off signal generator and Kepco power supply.
- Finish traveler - make sure all data is recorded.

DCCT Test Results for DCCT S/N: DCCT-M-B-0011

Technician: Mike, Life #: 26357

Test performed: 03/16/25, 08:41 PM

Raw Serial Number/Sticker Data: DCCT-M-B-0011-70000000006-BZ20130002-HBA20030148-22211030016-BZ22020423-HBA22170043-CTP-6303-50-A

FLT12 Fault Signal 1 and 2 Test Results

Test	Value	Status
FLT12 Overall Testing	N/A	Pass
FLT12 Positive Assertion	N/A	Pass
FLT12 Negative Assertion	N/A	Pass
FLT12 De-Assertion	N/A	Pass
FLT12 Initial Voltage	14.77 V	Pass
FLT12 Positive Fault Voltage	-0.00 V	Pass
FLT12 Negative Fault Voltage	-0.00 V	Pass

Pass/Fail Threshold Voltage for FLT12 Initial Voltage: >14.3V

Pass/Fail Threshold Voltage for FLT12 Fault Voltage: 1V

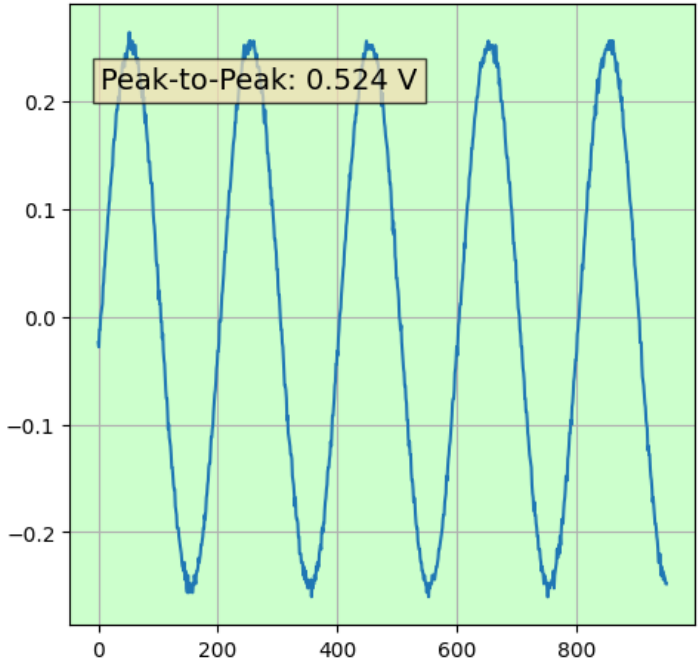
FLT12 Fault testing carried out with +/-1A current through DCCT

Test Current for IOUT Measurement: +/-10A at 10Hz Sinusoidal
20Vpp sine into Kepco current control input)

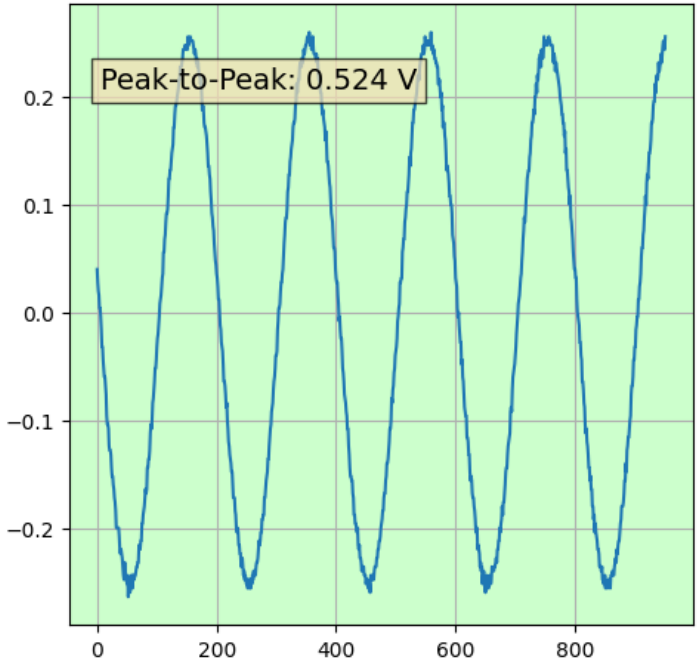
Current Test Waveforms

ALSu DCCT Test Data

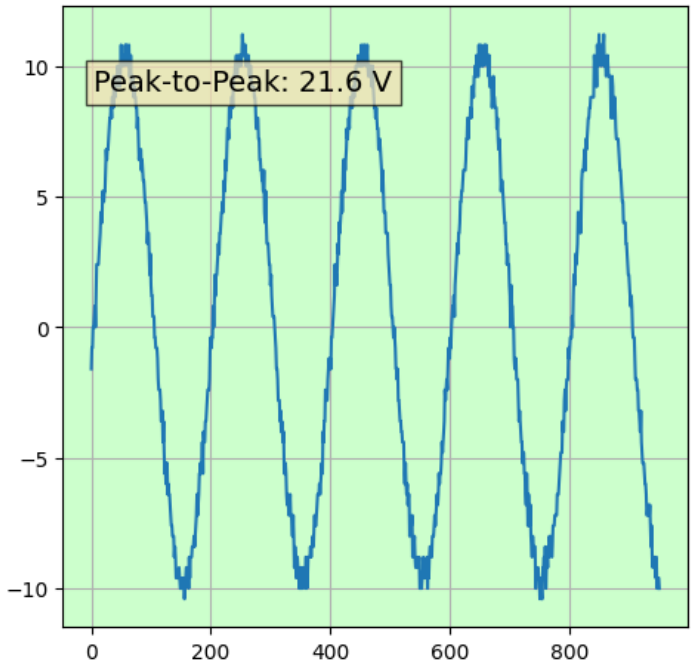
Channel 1, IOUT 1



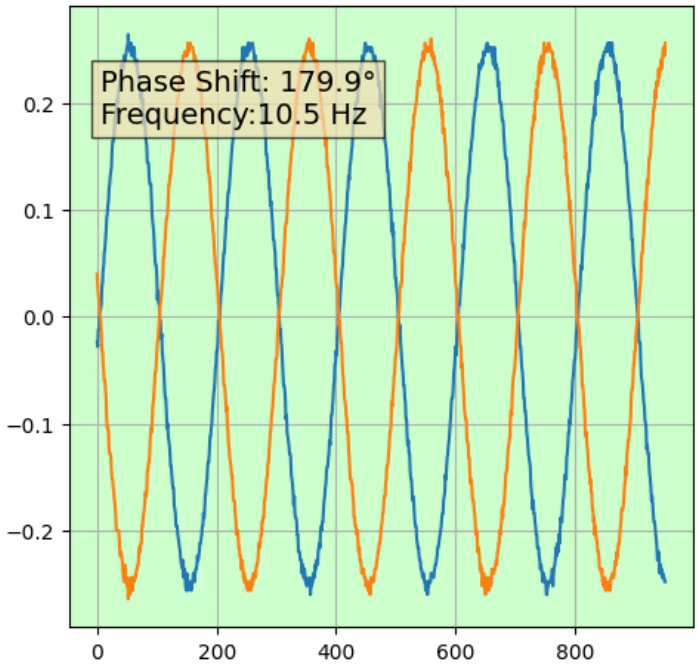
Channel 2, IOUT 2



Channel 3, SIGNAL GEN



Channel 1 & Channel 2 (IOUT 1 and IOUT 2) Phase Shift



DCCT Test Results for DCCT S/N: DCCT-M-B-0011

Technician: Mike, Life #: 26357

Test performed: 03/16/25, 08:45 PM

Raw Serial Number/Sticker Data: DCCT-M-B-0011-70000000006-BZ20130002-HBA20030148-22211030016-BZ22020423-HBA22170043-CTP-6303-50-A

FLT12 Fault Signal 1 and 2 Test Results

Test	Value	Status
FLT12 Overall Testing	N/A	Pass
FLT12 Positive Assertion	N/A	Pass
FLT12 Negative Assertion	N/A	Pass
FLT12 De-Assertion	N/A	Pass
FLT12 Initial Voltage	14.77 V	Pass
FLT12 Positive Fault Voltage	-0.00 V	Pass
FLT12 Negative Fault Voltage	-0.00 V	Pass

Pass/Fail Threshold Voltage for FLT12 Initial Voltage: >14.3V

Pass/Fail Threshold Voltage for FLT12 Fault Voltage: 1V

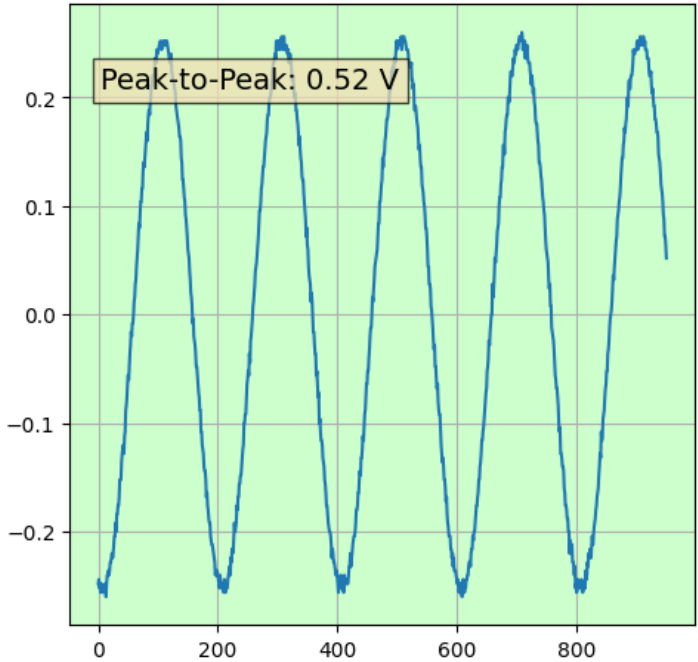
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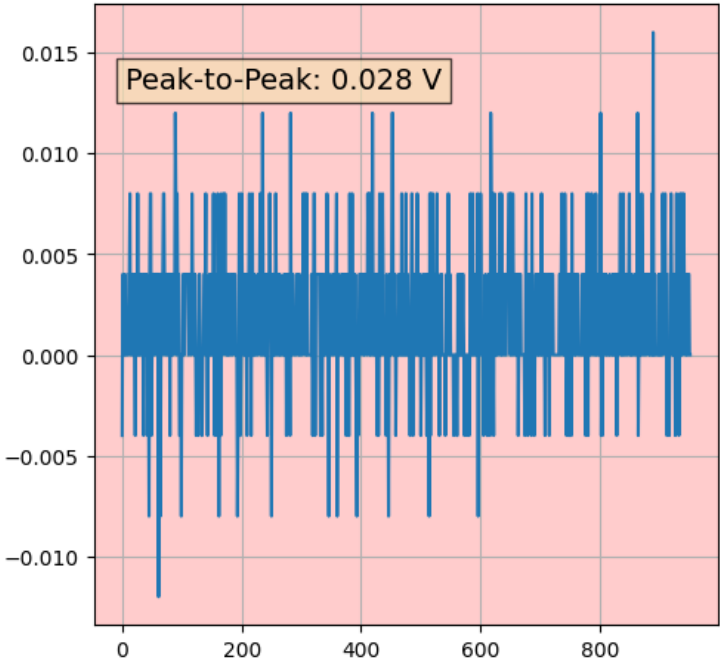
Current Test Waveforms

ALSu DCCT Test Data

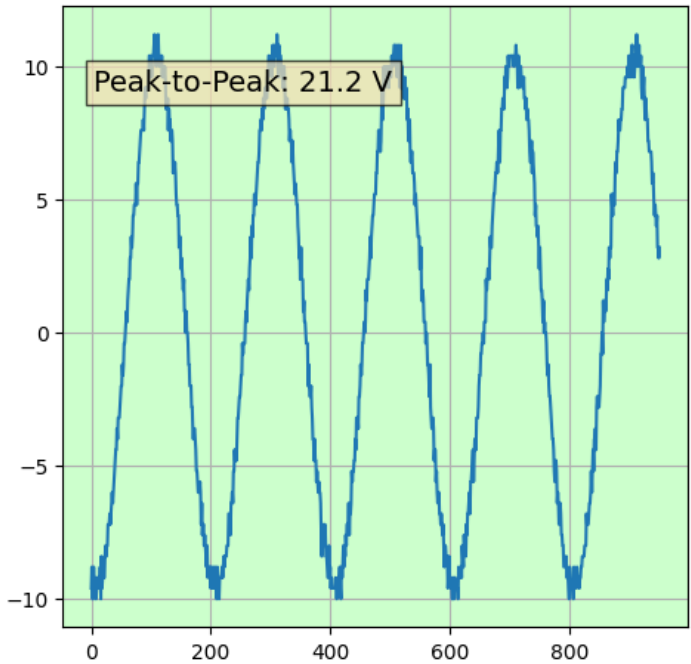
Channel 1, IOUT 1



Channel 2, IOUT 2



Channel 3, SIGNAL GEN



Channel 1 & Channel 2 (IOUT 1 and IOUT 2) Phase Shift

